A3-PS003-TC092

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

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Problem Description

Problem Statement

- A Centralized traffic-flow prediction can require continuous data transmission to a central server, increasing communication overhead.
- Federated Learning enables multiple clients to collaboratively train a model locally without sharing raw training data to cloud for privacy reasons.

Key Challenges

- Communication with cloud for exchanging model updates.
- Data Heterogeneity: Handling different traffic patterns across intersections.
- Prediction Performance: Evaluating accuracy and convergence under IID and Non-IID data distributions.

Background

- **Traffic-Flow Prediction:** Uses historical traffic data to estimate future traffic conditions.
- **Federated Learning (FL):** Enables multiple clients to collaboratively train a shared model without directly exchanging their raw traffic data.
- **Federated Averaging (FedAvg):** Allows clients to train models locally and send model weight updates to a central server, which aggregates them to create an updated global model at cloud.
- **IID and Non-IID Data:** Traffic observations may be similarly distributed across clients (IID) or differ significantly between intersections (Non-IID), potentially affecting model performance.
- **Proposed solutions:**
- Adaptive Local Training: Each client choose its local training effort based on its traffic-data characteristics.
- Communication-Efficient FedAvg: sending the model update every round regardless of whether it changed significantly, introduce an update threshold.

Literature & RFC Review

1. Federated Learning — McMahan et al. (2017)

- Introduced Federated Averaging (FedAvg), enabling multiple clients to train a shared model without exchanging raw training data.
- Relevance: Provides the fundamental algorithm for collaborative traffic-flow prediction in our project.

2. Machine Learning Approach for Traffic Flow Prediction - Mariaelena Berlotti , Sarah Di Grande, Salvatore Cavalieri

- Traditional centralized approaches collect data at a central server to train a prediction model.
- The pridiction is divided into two levels, first level uses an unsupervised clustering model to extract patterns from sensor-generated data, while the second level employs supervised machine learning models.
- Limitation: the proposed approach requires the availability of huge data from traffic sensors to realize the training of the machine learning models.

Proposed Solution

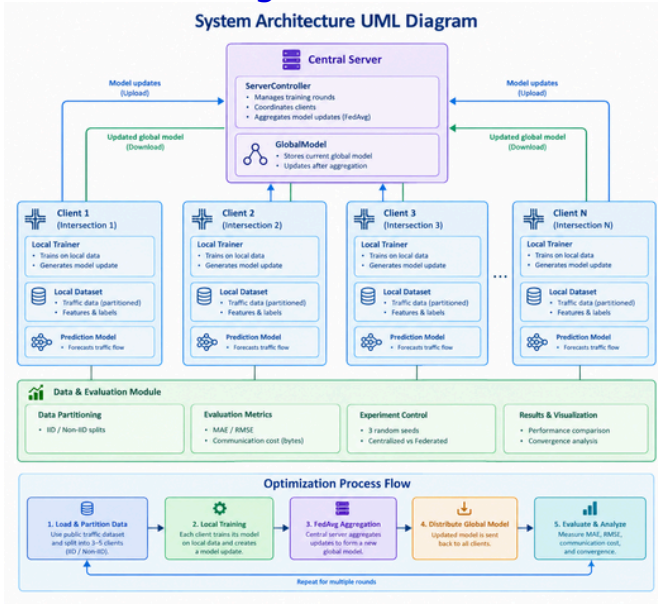
Overall Solution

- Develop a traffic-flow prediction system using Federated Learning (FL) and the FedAvg algorithm.
- Simulate 3–5 clients, each representing a traffic intersection with its own local dataset.
- Train a prediction model locally at each client without sharing raw traffic data.
- Send model updates to a central server, which aggregates them to create an improved global model.
- Adaptive Local Training: Each client chooses its local training effort based on its traffic-data characteristics.
- Communication-Efficient FedAvg: sending the model update every round regardless of whether it is significant or not which consumes bandwidth, so an update threshold is introduced.
- Distribute the updated global model back to clients and repeat the process across multiple training rounds.

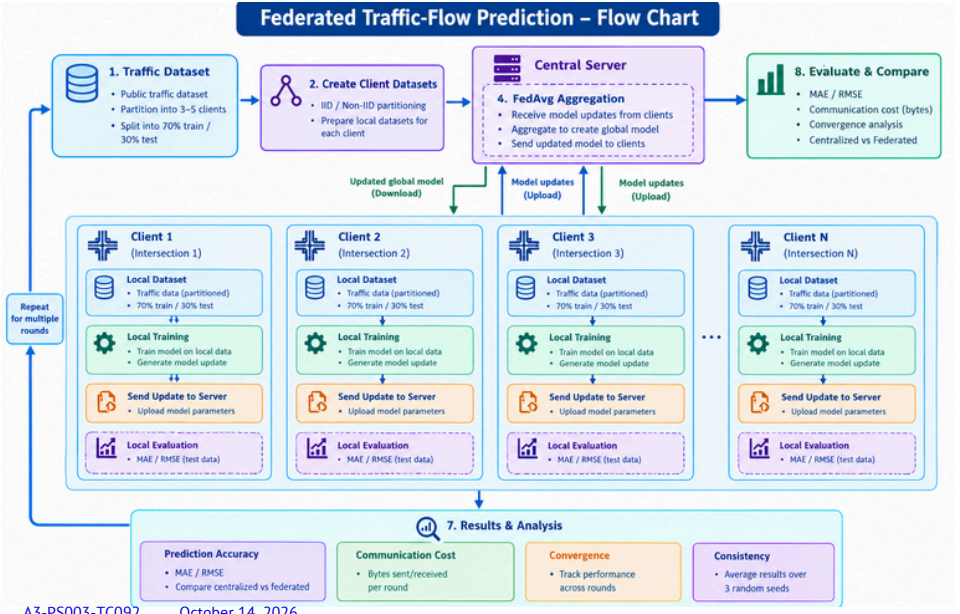
Optimization Proposed

Area	Before: Basic Approach	After: Proposed Enhancements
Communication Cost	Model updates exchanged every training round.	Model updates exchange when the update is significant.
Client Data Distribution	Training with one data distribution setup.	Compare IID and Non-IID traffic data across clients.
Model Evaluation	Evaluate prediction performance only.	Evaluate both MAE/RMSE and communication overhead.
Training at edge device	Fixed training i.e fixed number of epoches	Adaptive training i.e epoches change based on the traffic characteristics
Performance Comparison	Evaluate federated learning independently.	Compare centralized and federated learning under comparable conditions.

Solution Architecture & Design



Data Flow Diagram



Deployment Timeline

Milestone	Key Activities	Expected Deliverable
Data Preparation	Select a public traffic dataset, preprocess data, split into 70% training and 30% testing data.	Prepared Dataset
Model Development	Implement centralized training, simulate 3–5 clients, and implement federated learning using FedAvg.	Working prediction and FL Models
Optimization and Testing	Test IID and Non-IID distributions, measure communication cost, and evaluate MAE/RMSE.	Experimental Results
Evaluation & Finalization	Repeat experiments with three random seeds, compare centralized and federated results, and prepare the final report.	Performance comparison and final presentation

Bibliography I

[1] Exploration and Analysis of FedAvg - Jinlin Li ,Swjtu-Leeds Joint School, Southwest Jiaotong University, Chengdu, Sichuan, 611756, China

[2] Machine learning based adaptive traffic prediction and control using edge impulse platform - Manoj Tolani, G. E. Saathwik, Ayush Roy, L. A. Ameeth, Dhanush Bharadwaj Rao, Ambar Bajpai & Arun Balodi, Article number: 17161 (2025)

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Thank You for Your Attention!